

Multiple Choice (10 marks)

1. Let $g(x)$ be the inverse of $f(x) = x + \cos(x)$. What is $g'(1)$?
 - (a) $\sqrt{2}$
 - (b) -2
 - (c) 1
 - (d) 0
 - (e) $-1/2$
2. Find the degree for the given field extension $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{18})$ over $\mathbb{Q}$.
 - (a) 3
 - (b) 10
 - (c) 0
 - (d) 12
 - (e) 5
3. In a group of 10 scores, the largest score is increased by 40 points. What will happen to the mean?
 - (a) It will remain the same.
 - (b) It will increase by 30 points.
 - (c) It will increase by 10 points.
 - (d) It will decrease by 10 points.
 - (e) It will remain unchanged if the other scores also increase by 40 points.
4. What is the second number in the row of Pascal's triangle that has 43 numbers?
 - (a) 42
 - (b) 1
 - (c) 2
 - (d) 40
 - (e) 22
5. Let $P_5(x)$ be the fifth-degree Taylor polynomial approximation for $f(x)=\sin(x)$, centered at $x=0$. What is the Lagrange error of the polynomial approximation to $\sin(1)$?
 - (a) 0.02098
 - (b) 0.00198

- (c) 0.000198
- (d) 0.01998
- (e) 0.0198

True / False (10 marks)

Answer True or False

6. True or False: Rob uses approximately 6 boxes of cat food in a month, given he consumes 1 box every 5 days.
7. True or False: There are 25000 valid paths from (0,0) to (10,10) on a grid that do not cross the diagonal line $y = x$.
8. True or False: The number 640, when divided by 4, yields a quotient of 160 and a remainder of 0.
9. True or False: The determinant of matrix $A = \begin{bmatrix} 2 & 4 & 3 \\ 3 & 0 & 1 \\ 1 & 2 & 5 \end{bmatrix}$ is equal to 30.
10. True or False: The expressions $4n + 2$ and $8n$ are equivalent for any variable values greater than zero.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of rounding in mathematical contexts by identifying possible actual values of runners that would round to 400. Analyze how rounding affects data interpretation.
12. Analyze the structure of the symmetric group S_5 and determine the largest order of an element within this group. Explain your reasoning and the implications of your findings.

End of Paper